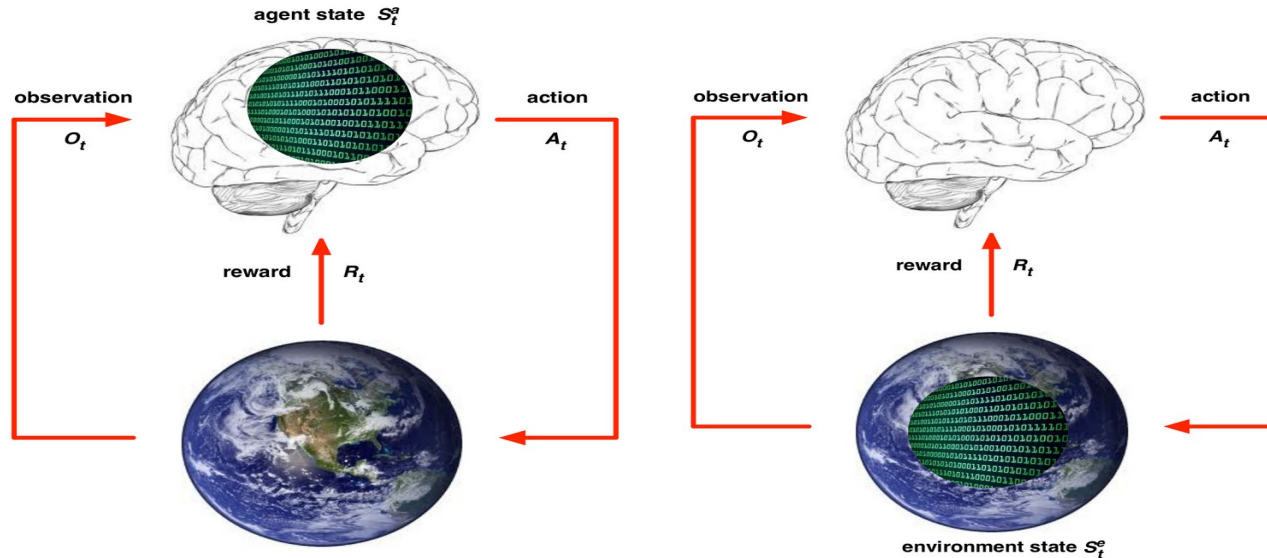


Fully Observable Environments



Full observability: agent directly observes environment state

$$O_t = S_t^a = S_t^e$$

Agent state = environment state = information state

Formally, this is a Markov decision process (MDP)

Partially Observable Environments

Partial observability: agent indirectly observes environment:

- A robot with camera vision isn't told its absolute location

- A trading agent only observes current prices

- A poker playing agent only observes public cards

Now agent state NOT EQUAL TO environment state

Formally this is a partially observable Markov decision process (POMDP)

Agent must construct its own state representation S^a

- Complete history: $S_t^a = H_t$

- Beliefs** of environment state: $S_t^a = (\mathbb{P}[S_t^e = s^1], \dots, \mathbb{P}[S_t^e = s^n])$

- Recurrent neural network: $S_t^a = \sigma(S_{t-1}^a W_s + O_t W_o)$

Major Components of an RL Agent

An RL agent may include one or more of these components:

- Policy: agent's behaviour function
- Value function: how good is each state and/or action
- Model: agent's representation of the environment

Policy

- A policy is the agent's behaviour
- Policy π determines how the agent chooses actions
- It is a map from state to action ($\pi : S \rightarrow A$), e.g.
 - Deterministic policy: $a = \pi(s)$
 - Stochastic policy: $\pi(a|s) = P[A_t = a|S_t = s]$

Value Function

- Value function is a prediction of future reward
- Used to evaluate the goodness/badness of states
- And therefore to select between actions, e.g.

$$v_{\pi}(s) = \mathbb{E}_{\pi} [R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \mid S_t = s]$$

A Model

- A model predicts what the environment will do next
- P predicts the next state
- R predicts the next (immediate) reward, e.g.

$$\mathcal{P}_{ss'}^a = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a]$$

$$\mathcal{R}_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$$

Evaluation and Control

- Evaluation
 - Estimate/predict the expected rewards from following a given policy
- Control
 - Optimization: find the best policy

Making Sequences of Good Decisions Given a Model of the World

- Assume finite set of states and actions
- Given models of the world (dynamics and reward)
- Evaluate the performance of a particular decision policy
- Compute the best policy
- This can be viewed as an AI planning problem

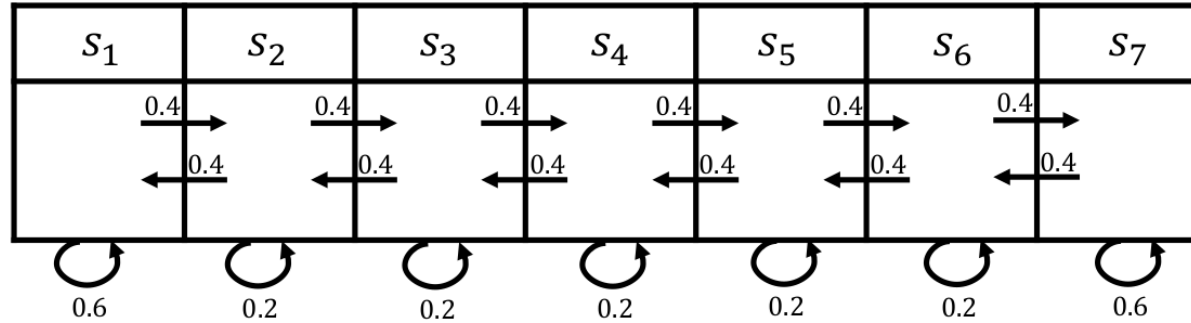
Making Sequences of Good Decisions Given a Model of the World

- Markov Processes
- Markov Reward Processes (MRPs)
- Markov Decision Processes (MDPs)
- Evaluation and Control in MDPs

Markov Process or Markov Chain

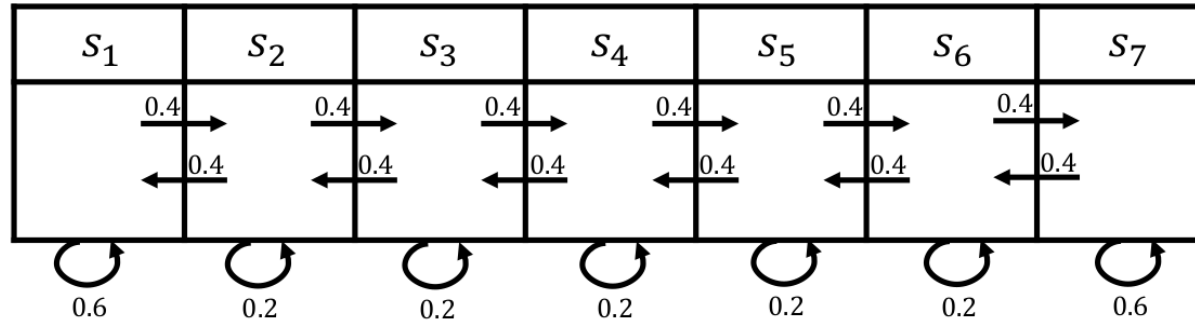
- Memoryless random process
 - Sequence of random states with Markov property
- Definition of Markov Process
 - S is a (finite) set of states ($s \in S$)
 - P is dynamics/transition model that specifies $P(s_{t+1} = s' | s_t = s)$
- Note: no rewards, no actions
- If finite number (N) of states, can express P as a matrix

Markov Chain Transition Matrix



$$P = \begin{pmatrix} 0.6 & 0.4 & 0 & 0 & 0 & 0 & 0 \\ 0.4 & 0.2 & 0.4 & 0 & 0 & 0 & 0 \\ 0 & 0.4 & 0.2 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.4 & 0.2 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0.4 & 0.2 & 0.4 & 0 \\ 0 & 0 & 0 & 0 & 0.4 & 0.2 & 0.4 \\ 0 & 0 & 0 & 0 & 0 & 0.4 & 0.6 \end{pmatrix}$$

Markov Chain Episodes



Example: Sample episodes starting from s_4

$s_4, s_5, s_6, s_7, s_7, s_7, \dots$

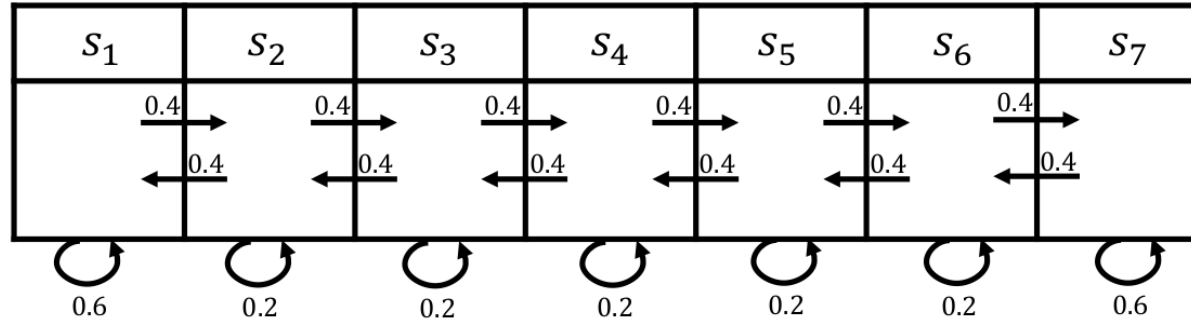
$s_4, s_4, s_5, s_4, s_5, s_6, \dots$

$s_4, s_3, s_2, s_1, \dots$

Markov Reward Process (MRP)

- Markov Reward Process is a Markov Chain + rewards
- Definition of Markov Reward Process (MRP)
 - S is a (finite) set of states ($s \in S$)
 - P is dynamics/transition model that specifies $P(s_{t+1} = s' | s_t = s)$
 - R is a reward function $R(s_t = s) = E[r_t | s_t = s]$
 - Discount factor $\gamma \in [0, 1]$
- Note: no actions
- If finite number (N) of states, can express R as a vector

Markov Chain Transition Matrix



- Reward: +1 in s_1 , +10 in s_7 , 0 in all other states

Return & Value Function

- Definition of Horizon (H)
 - Number of time steps in each episode
 - Can be infinite
 - Otherwise called finite Markov reward process
- Definition of Return, G_t (for a Markov Reward Process)
 - Discounted sum of rewards from time step t to horizon H
$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^{H-1} r_{t+H-1}$$
- Definition of State Value Function, $V(s)$ (for a Markov Reward Process)
 - Expected return from starting in state s

$$V(s) = \mathbb{E}[G_t | s_t = s] = \mathbb{E}[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^{H-1} r_{t+H-1} | s_t = s]$$

Discount Factor

- Mathematically convenient (avoid infinite returns and values)
- Humans often act as if there's a discount factor < 1
- $\gamma = 0$: Only care about immediate reward
- $\gamma = 1$: Future reward is as beneficial as immediate reward
- If episode lengths are always finite ($H < \infty$), can use $\gamma = 1$

Computing the Value of a Markov Reward Process

- Markov property provides structure
- MRP value function satisfies

$$V(s) = \underbrace{R(s)}_{\text{Immediate reward}} + \underbrace{\gamma \sum_{s' \in S} P(s'|s) V(s')}_{\text{Discounted sum of future rewards}}$$

Categorizing RL agents

- Value Based
 - Value Function
- Policy Based
 - Policy
- Actor Critic
 - Policy
 - Value Function

Model Free

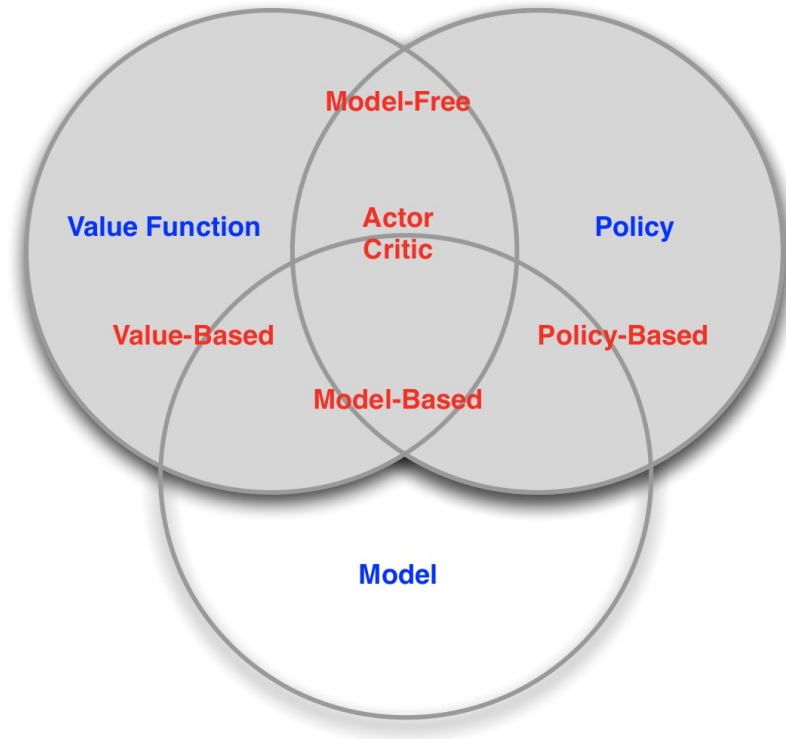
Policy and/or Value Function

Model Based

Policy and/or Value Function

Model

RL Agent Taxonomy



Learning and Planning

Two fundamental problems in sequential decision making

- Reinforcement Learning:
 - The environment is initially unknown
 - The agent interacts with the environment
 - The agent improves its policy
- Planning:
 - A model of the environment is known
 - The agent performs computations with its model (without any external interaction)
 - The agent improves its policy
 - a.k.a. deliberation, reasoning, introspection, pondering, thought, search

Question?

- Is it possible to play a game without knowing the rules of the game?

Exploration and Exploitation

- Reinforcement learning is like trial-and-error learning
- The agent should discover a good policy
- From its experiences of the environment
- Without losing too much reward along the way
- Exploration finds more information about the environment
- Exploitation exploits known information to maximise reward
- It is usually important to explore as well as exploit

Prediction and Control

- Prediction: evaluate the future
 - Given a policy
- Control: optimise the future
 - Find the best policy

Markov Process

- A Markov process is a memoryless random process, i.e. a sequence of random states $S_1, S_2, \dots$ with the Markov property.

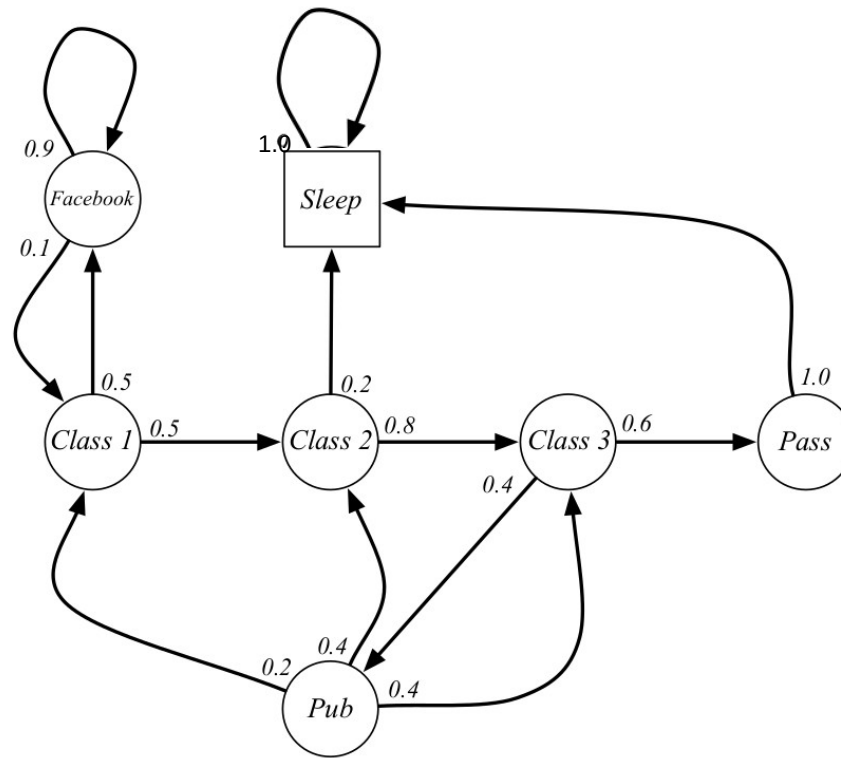
A Markov Process (or Markov Chain) is a tuple $\langle S, P \rangle$

S is a (finite) set of states

P is a state transition probability matrix,

$$P_{ss'} = \Pr [S_{t+1} = s' \mid S_t = s]$$

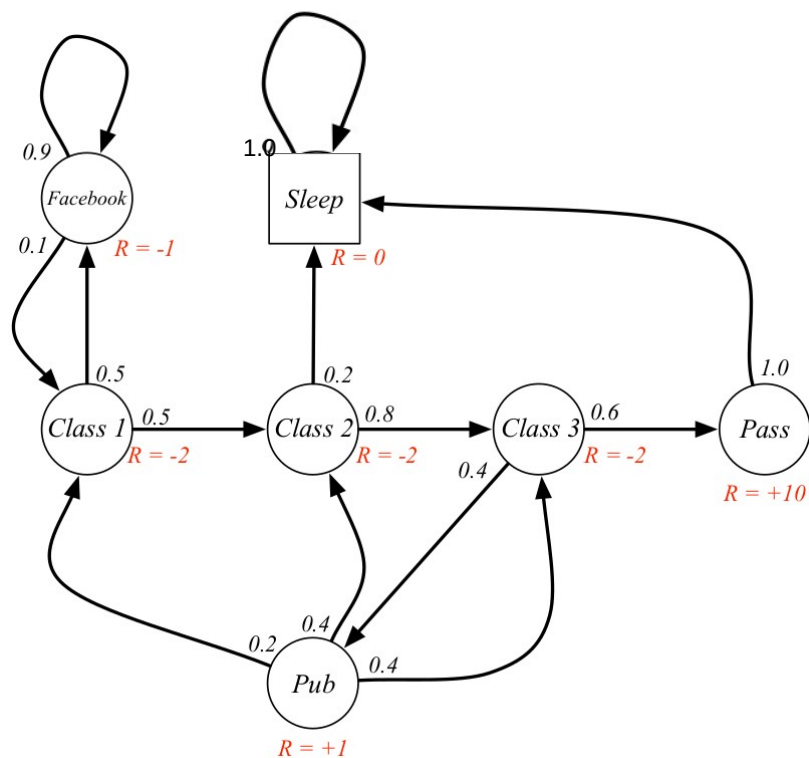
Student Markov Chain



Markov Reward Process

- A Markov reward process is a Markov chain with values.
- A *Markov Reward Process* is a tuple $\langle S, P, R, \gamma \rangle$
 - S is a finite set of states
 - P is a state transition probability matrix,
 - $P_{ss'} = \Pr[S_{t+1} = s' \mid S_t = s]$
 - R is a reward function, $R_s = E [R_{t+1} \mid S_t = s]$
 - γ is a discount factor, $\gamma \in [0, 1]$

Student MRP



Return

- The return G_t is the total discounted reward from time-step t .

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

The *discount* $\gamma \in [0, 1]$ is the present value of future rewards

The value of receiving reward R after $k + 1$ time-steps is $\gamma^k R$.

This values immediate reward above delayed reward.

Why discount?

- Most Markov reward and decision processes are discounted. Why?

Mathematically convenient to discount rewards

Avoids infinite returns in cyclic Markov processes

Uncertainty about the future may not be fully represented

If the reward is financial, immediate rewards may earn more interest than delayed rewards

Animal/human behaviour shows preference for immediate reward

It is sometimes possible to use *undiscounted* Markov reward processes (i.e. $\gamma = 1$), e.g. if all sequences terminate.

Value Function

- The value function $v(s)$ gives the long-term value of state s .

Definition: The state value function $v(s)$ of an MRP is the expected return starting from state s

$$v(s) = E[G_t | S_t = s]$$

Student MRP Returns

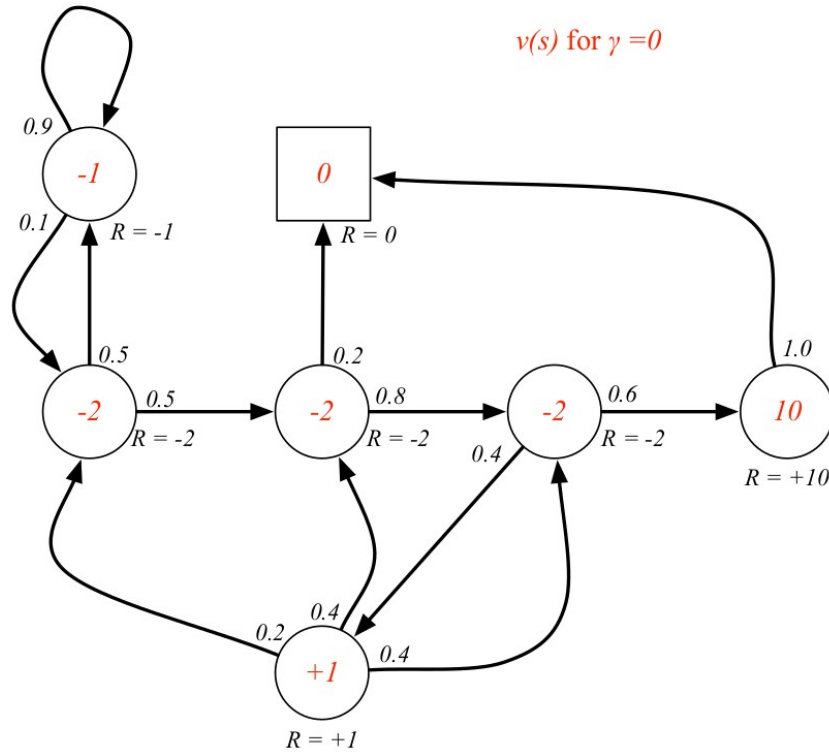
Sample **returns** for Student MRP:

Starting from $S_1 = C1$ with $\gamma = \frac{1}{2}$

$$G_1 = R_2 + \gamma R_3 + \dots + \gamma^{T-2} R_T$$

C1 C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 10 * \frac{1}{8}$	=	-2.25
C1 FB FB C1 C2 Sleep	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16}$	=	-3.125
C1 C2 C3 Pub C2 C3 Pass Sleep	$v_1 = -2 - 2 * \frac{1}{2} - 2 * \frac{1}{4} + 1 * \frac{1}{8} - 2 * \frac{1}{16} \dots$	=	-3.41
C1 FB FB C1 C2 C3 Pub C1 ...	$v_1 = -2 - 1 * \frac{1}{2} - 1 * \frac{1}{4} - 2 * \frac{1}{8} - 2 * \frac{1}{16} \dots$	=	-3.20
FB FB FB C1 C2 C3 Pub C2 Sleep			

State-Value Function for Student MRP



$$G_1 = R_2 + \gamma R_3 + \dots + \gamma^{T-2} R_T$$

Bellman Equation for MRP

- The value function can be decomposed into two parts:
 - immediate reward R_{t+1}
 - discounted value of successor state $\gamma v(S_{t+1})$

$$\begin{aligned}v(s) &= \mathbb{E} [G_t \mid S_t = s] \\&= \mathbb{E} [R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \mid S_t = s] \\&= \mathbb{E} [R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \dots) \mid S_t = s] \\&= \mathbb{E} [R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\&= \mathbb{E} [R_{t+1} + \gamma v(S_{t+1}) \mid S_t = s]\end{aligned}$$

Bellman Equation in Matrix Form

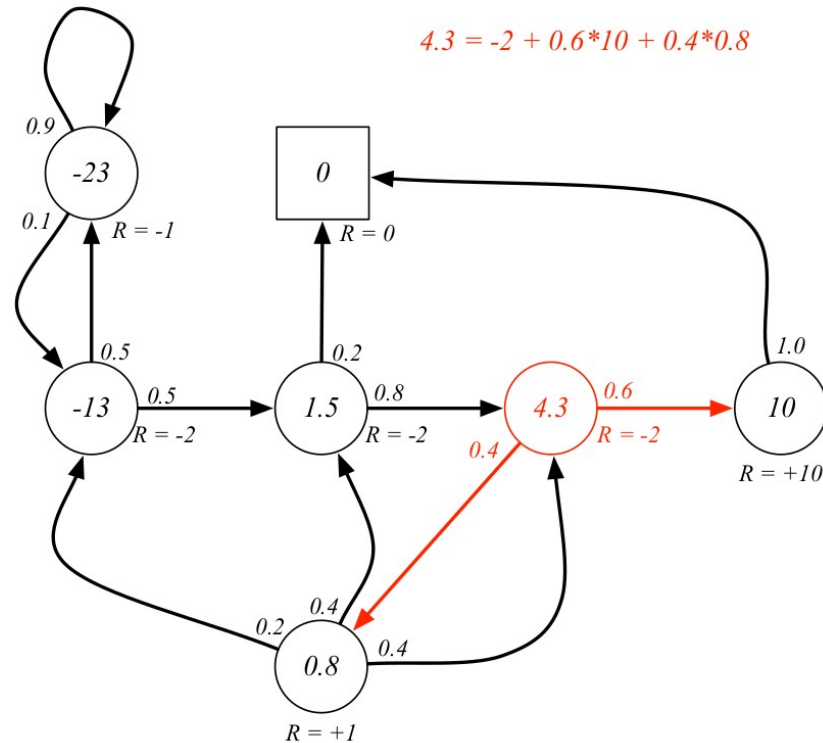
The Bellman equation can be expressed concisely using matrices,

$$v = \mathcal{R} + \gamma \mathcal{P}v$$

where v is a column vector with one entry per state

$$\begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix} = \begin{bmatrix} \mathcal{R}_1 \\ \vdots \\ \mathcal{R}_n \end{bmatrix} + \gamma \begin{bmatrix} \mathcal{P}_{11} & \dots & \mathcal{P}_{1n} \\ \vdots & & \\ \mathcal{P}_{n1} & \dots & \mathcal{P}_{nn} \end{bmatrix} \begin{bmatrix} v(1) \\ \vdots \\ v(n) \end{bmatrix}$$

Bellman Equation for Student MRP



Dynamic Programming

- Finding the shortest path in a graph using optimal substructure
- Mathematically, DP usually refers to simplifying a decision by breaking it down into a sequence of decision steps over time.
- Defined by a sequence of value functions $V_1, V_2, \dots, V_n$ taking y as an argument representing the state of the system at times i from 1 to n .
- In Bioinformatics
 - Sequence alignment, protein folding
- In control theory
 - Admissible control, admissible trajectory, minimizing the cost function.

Stochastic Dynamic Programming

- Used to modeling and solving decision making problems under uncertainty.
- Components:
 - An initial state
 - A decision variable (feasible action)
 - An immediate cost/reward function
 - A state transition function

Markov Decision Process

- A Markov decision process (MDP) is a Markov reward process with decisions. It is an environment in which all states are Markov.
- A Markov Decision Process is a tuple $\langle S, A, P, R, \gamma \rangle$

S is a finite set of states

A is a finite set of actions

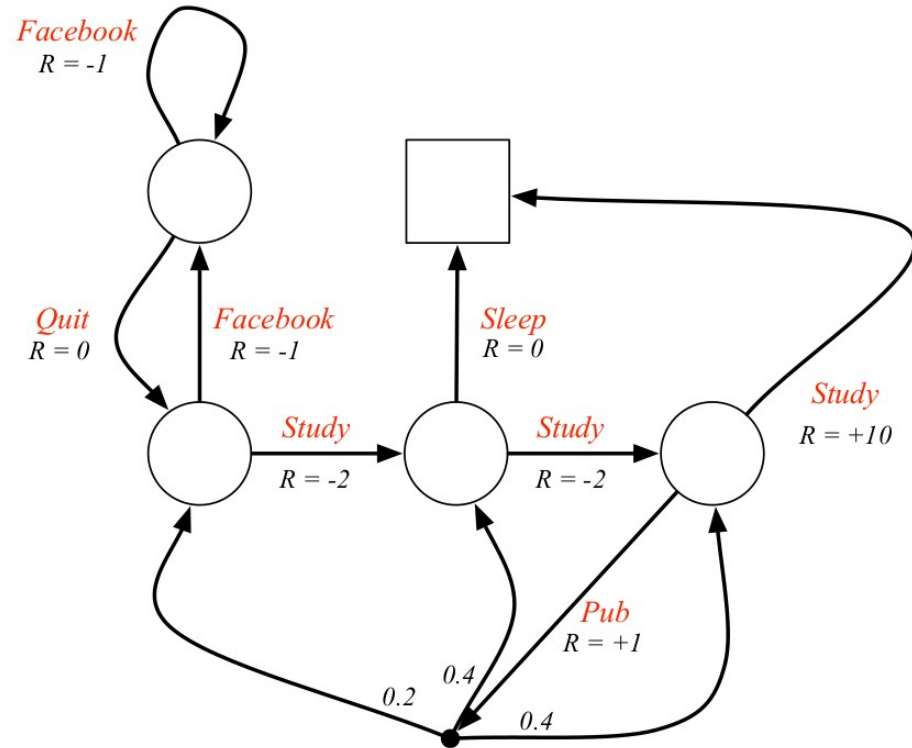
P is a state transition probability matrix,

$$P_{ss'}^a = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a]$$

R is a reward function, $R_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$

γ is a discount factor $\gamma \in [0, 1]$.

Student Markov Decision Process



Policies

- A policy π is a distribution over actions given states,

$$\pi(a|s) = P [A_t = a \mid S_t = s]$$

A policy fully defines the behaviour of an agent

MDP policies depend on the current state (not the history)

i.e. Policies are *stationary* (time-independent),

$$A_t \sim \pi(\cdot | S_t), \forall t > 0$$

Policies

Given an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a policy π

The state sequence $S_1, S_2, \dots$ is a Markov process $\langle \mathcal{S}, \mathcal{P}^\pi \rangle$

The state and reward sequence $S_1, R_1, S_2, \dots$ is a Markov reward process $\langle \mathcal{S}, \mathcal{P}^\pi, \mathcal{R}^\pi, \gamma \rangle$

where

$$\mathcal{P}_{s,s'}^\pi = \sum_{a \in \mathcal{A}} \pi(a|s) \mathcal{P}_{ss'}^a$$

$$\mathcal{R}_s^\pi = \sum_{a \in \mathcal{A}} \pi(a|s) \mathcal{R}_s^a$$

Value Function

The *state-value function* $v_{\pi}(s)$ of an MDP is the expected return starting from state s , and then following policy π

$$v_{\pi}(s) = E_{\pi}[G_t \mid S_t = s]$$

The *action-value function* $q_{\pi}(s, a)$ is the expected return starting from state s , taking action a , and then following policy π

$$q_{\pi}(s, a) = E_{\pi}[G_t \mid S_t = s, A_t = a]$$

State-Value Function for Student MDP

